

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Tuesday 6 January 2026

Afternoon (Time: 40 minutes)

Paper reference **31617H/1P**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Physics

SECTION C: WAVES IN COMMUNICATION

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
 - Section A: Structures and functions of cells and tissues (Biology)
 - Section B: Periodicity and properties of elements (Chemistry)
 - Section C: Waves in communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- The formulae sheet can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☐. If you change your mind about an answer, put a line through the box ☐ and then mark your new answer with a cross ☐.

- 1 (a) Figure 1 shows a displacement–distance graph for a wave.

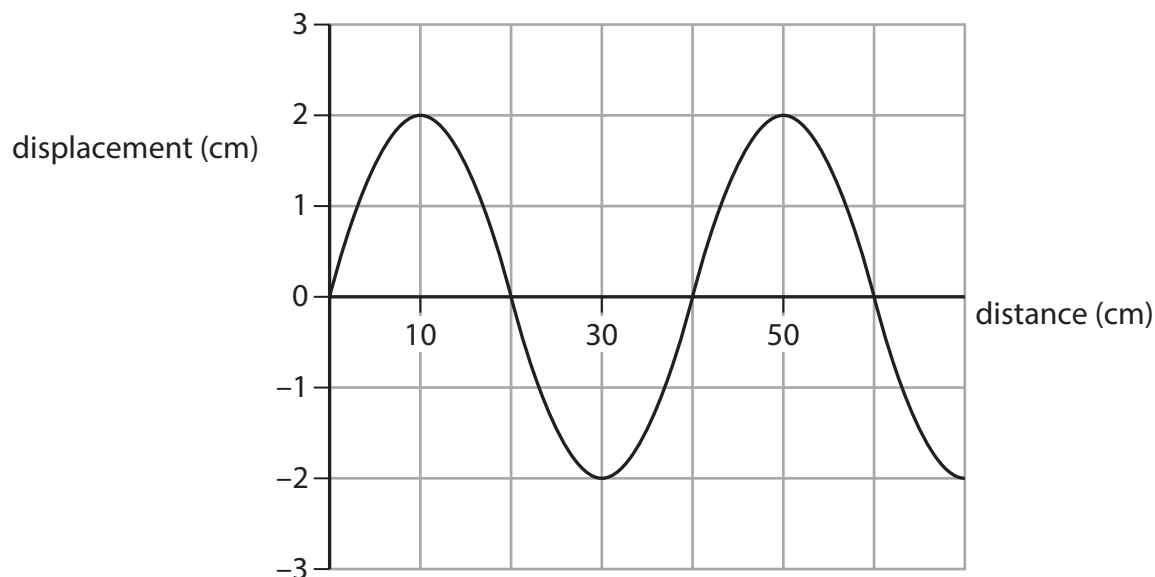


Figure 1

- (i) Identify the amplitude of the wave in Figure 1.

(1)

- ☐ **A** 2 cm
☐ **B** 4 cm
☐ **C** 20 cm
☐ **D** 40 cm

- (ii) Determine the wavelength of the wave in Figure 1.

(1)

wavelength = cm



Figure 2 shows a displacement–distance graph for two waves.

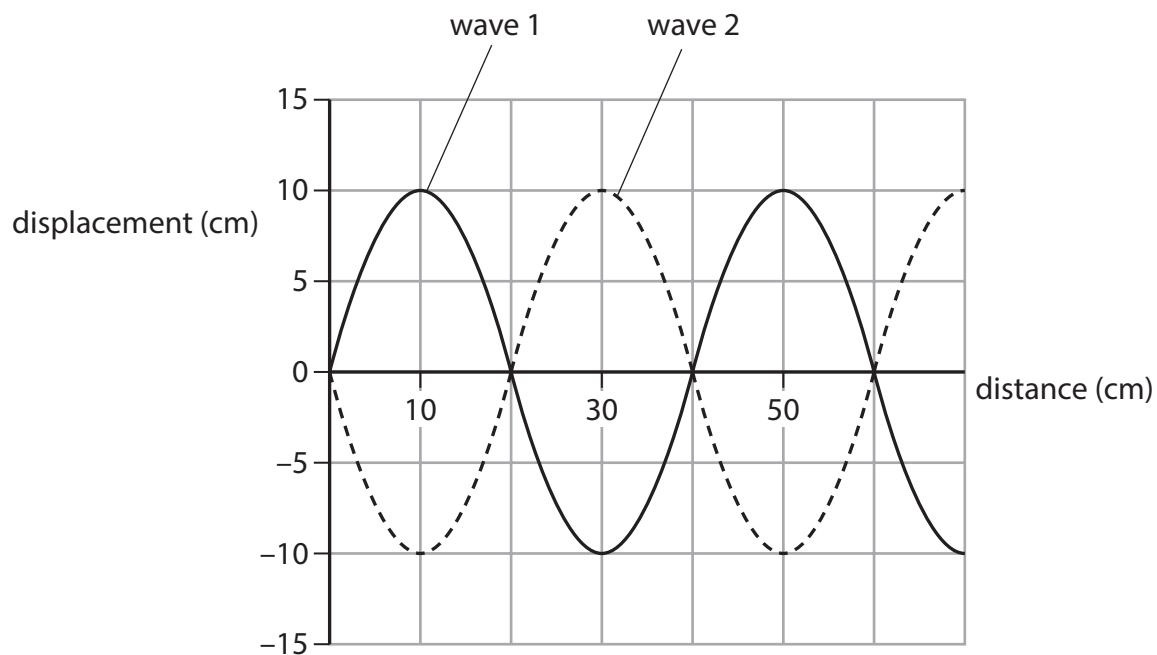


Figure 2

(b) Identify the phase difference between wave 1 and wave 2 in Figure 2.

(1)

- ☐ **A** 45°
- ☐ **B** 90°
- ☐ **C** 180°
- ☐ **D** 270°

The periodic time (T) of a wave is 4.0 ms.

(c) Calculate the frequency (f) of the wave.

(3)

Use the equation

$$f = \frac{1}{T}$$

Show your working.

frequency = Hz

(Total for Question 1 = 6 marks)



- 2 Figure 3 shows an incident wave and its reflected wave when a guitar string is plucked.

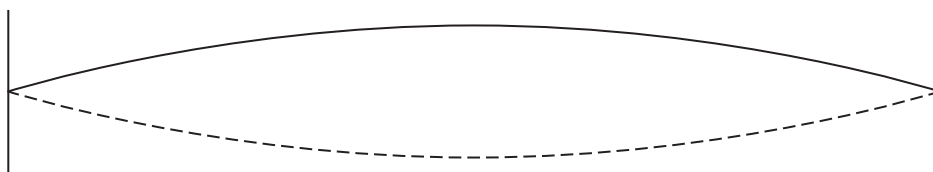


Figure 3

- (a) Identify the harmonic shown in Figure 3.

(1)

- ☐ A first
☐ B second
☐ C third
☐ D fourth

- (b) The tension in the guitar string is increased.

- (i) Explain how increasing the tension changes the frequency of the wave on the guitar string.

(3)

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The mass per unit length (μ) of another guitar string is $2.35 \times 10^{-3} \text{ kg m}^{-1}$.

The velocity (v) of a wave on the string is 158 m s^{-1} .

(ii) Calculate the tension (T) in the string.

(4)

Use the equation

$$v = \sqrt{\frac{T}{\mu}}$$

Show your working.

tension = N



Figure 4 shows another incident wave and its reflected wave when the guitar string is plucked.

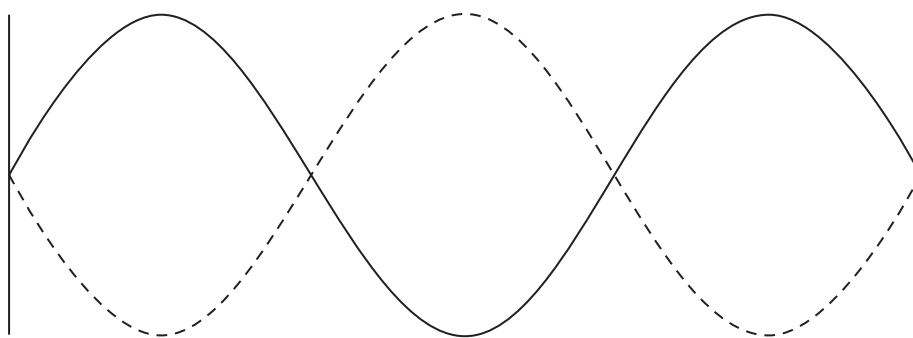


Figure 4

(c) Describe how these waves produce a standing wave.

(2)

(Total for Question 2 = 10 marks)



3 (a) Waves in the electromagnetic spectrum have many applications.

- (i) State the region of the electromagnetic spectrum that has waves with the highest frequency.

(1)

Radio waves are part of the electromagnetic spectrum that is used in Bluetooth® technology.

- (ii) Explain **one** feature of Bluetooth® technology that reduces interference from other devices that use radio waves.

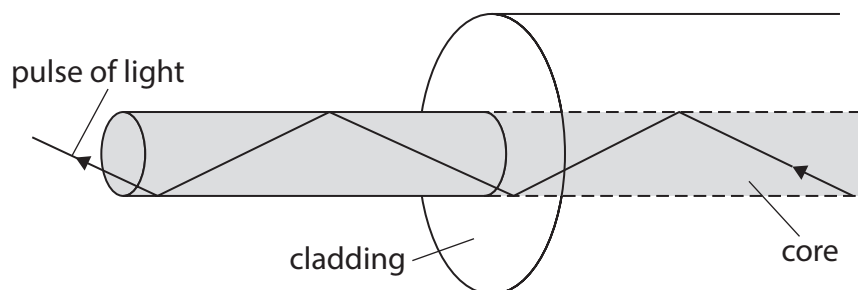
(2)



Visible light is a region of the electromagnetic spectrum used in communications.

Figure 5 shows an optical fibre.

A pulse of light travels down the optical fibre by total internal reflection.



(Source: ©PAL)

Figure 5

- (b) Explain how the optical fibre is designed for the total internal reflection of the pulse of light.

(3)

(Total for Question 3 = 6 marks)



4 Mobile phone networks use microwave signals.

(a) Describe how a signal is transmitted from one mobile phone to another.

(2)

Satellite communication uses radio waves and microwaves to transmit information.

Figure 6 shows how radio waves and microwaves are affected by the ionosphere.

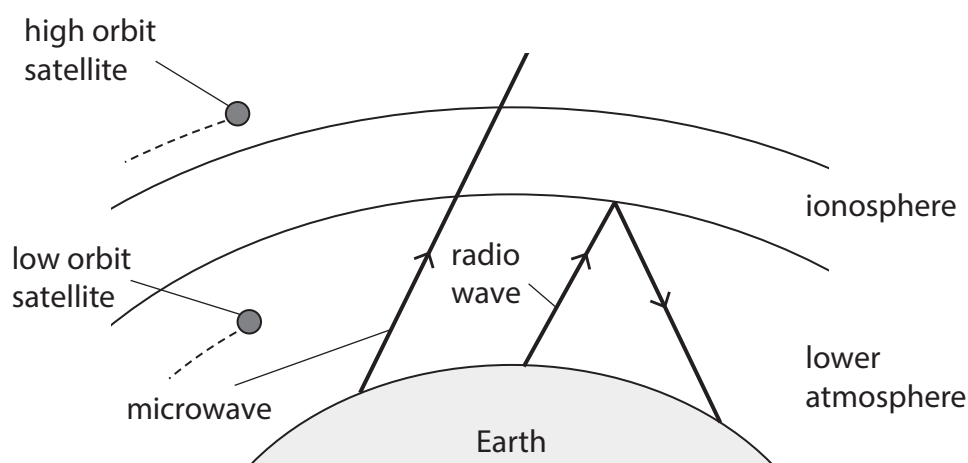


Figure 6

(b) Discuss the differences between satellite communication using radio waves and satellite communication using microwaves.

Use information from Figure 6 in your answer.

(6)



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(Total for Question 4 = 8 marks)

TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



Formulae sheet

Wave speed

$$v = f\lambda$$

Speed of a transverse wave on a string

$$v = \sqrt{\frac{T}{\mu}}$$

Refractive index

$$n = \frac{c}{v} = \frac{\sin i}{\sin r}$$

Critical angle

$$\sin C = \frac{1}{n}$$

Inverse square law in relation to the intensity of a wave

$$I = \frac{k}{r^2}$$

$$\frac{I_1}{I_2} = \frac{(D_2)^2}{(D_1)^2}$$

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